

Sahil Pal

Software Engineer

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Summary

Software Engineer specializing in AI systems, backend architecture, and developer tooling. Experienced designing production-grade Retrieval-Augmented Generation (RAG) pipelines, context engines for AI agents, and scalable REST APIs using Python, TypeScript, and vector databases such as pgvector and SQLite FTS5. Comfortable owning the full stack, from data ingestion to semantic retrieval and production deployment.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL

AI & Machine Learning: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, Vector Search, Semantic Embeddings, Model Context Protocol (MCP), Gemini API

Backend & Systems: Node.js, Express, Next.js, REST APIs, Server-Sent Events (SSE), Worker Queues, Async Pipelines, Event-Driven Architecture

Databases & Search: PostgreSQL, pgvector, SQLite, SQLite FTS5, MongoDB, Redis, Supabase, Vector Databases

Tools & Frameworks: React, Tailwind CSS, Vite, Tree-sitter, Git, GitHub Actions (CI/CD), Manifest V3, Docker

Projects

• Synapse

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Structural context engine for AI agents — Python, SQLite, MCP, Tree-sitter

- Architected a Git-aware daemon in Python that parses codebases into AST symbols and dependency graphs, persisting state in a Write-Ahead Logging (WAL) SQLite database for concurrent, crash-safe access.
- Built a Model Context Protocol (MCP) server that serves context-bounded code summaries and semantically-ranked code snippets to AI coding agents in real time.
- Engineered an asynchronous (`asyncio`) background worker to generate lazy-loaded semantic Markdown documentation for files and modules without blocking the main process.
- Designed a hybrid retrieval engine combining vector similarity search with SQLite FTS5 keyword search over code embeddings, improving contextual relevance for agent queries.

• RAG Doc Analyzer

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Document Q&A AI Platform — Next.js, Express, PostgreSQL, pgvector, Gemini

- Designed a grounded RAG pipeline combining Google Gemini embeddings, pgvector similarity search, and PostgreSQL full-text fallback, streamed to the client via Server-Sent Events (SSE).
- Built a chat-first, multi-document Q&A platform with Firebase authentication, persistent user workspaces, and cross-document context orchestration.
- Engineered an asynchronous document ingestion and chunking pipeline using a PostgreSQL worker queue (`FOR UPDATE SKIP LOCKED`) for safe concurrent processing, with file storage on Supabase.

• LeetSync

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Chrome Extension & Sync Platform — TypeScript, Vite, Manifest V3

- Developed a Manifest V3 Chrome extension that captures LeetCode editor content and problem metadata, converting submissions into structured Markdown notes.
- Implemented in-page context injection (popup UI + Shadow DOM isolation) for seamless, non-intrusive capture of code and notes within the browser tab.
- Automated version-controlled synchronization to GitHub via the Contents API, organizing solutions into structured repositories and eliminating manual commits.

Experience

• Software Engineering Intern

June 2025 – July 2025

Kristu Jayanti Software Development Centre — Kristu Jayanti University

- Developed RESTful APIs in Java 17 with Vert.x for authentication, user profiles, item management, search, and content moderation.
- Implemented production-grade authentication and authorization, including JWT-based sessions, email verification, password reset flows, Redis token blacklisting, and role-based access control (RBAC).
- Designed MongoDB-backed workflows integrating SMTP notifications, scheduled cleanup jobs, and asynchronous handlers to coordinate database, cache, and email systems reliably.

Education

• Bachelor of Computer Applications (BCA)

2023 – 2026

Kristu Jayanti College, Bengaluru

CGPA: 7.9/10